Griffin Staples

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Marine robotics · Field deployment · Edge ML · Software development

Profile

I'm an MASc candidate in Electrical and Computer Engineering at UBC, where I design, build and deploy robotic systems, from an underwater ROV fielded with the Sc'ianew First Nation to a bio-inspired surface vehicle. I bring nearly five years of professional software development experience, including ISS centrifuge control software at MDA. I'm seeking hands-on robotics roles and am open to field work.

Skills & Certifications

- **Field & robotic systems:** ROV and surface-vehicle build, deployment and testing; vehicle simulation; flight avionics; real-time telemetry and data acquisition
- **Field certifications:** ROC(M) VHF marine radio · Pleasure Craft Operator Card · Wilderness First Aid
- **Embedded & hardware:** Arduino, ESP32, Onshape CAD, prototyping and fabrication
- **Machine learning:** deep reinforcement learning, explainable ML, model development and evaluation, LLM/RAG applications
- **Software:** Python, C++, TypeScript, JavaScript, Linux, Git, Google Cloud (Pub/Sub), test infrastructure

Education

Master of Applied Science, Electrical and Computer Engineering — University of British Columbia, Vancouver, BC | Sept 2025 – Present

- Marine Robotics Lab, supervised by Dr. Adrien Desjardins · Cumulative average: 91.3%

Bachelor of Engineering, Engineering Physics — Queen's University, Kingston, ON | 2015 – 2020

- GPA: 3.86 · Dean's Scholar, 2015–2020
- Design thesis: designed and prototyped a lightweight, low-cost ambulatory blood pressure monitor; trained an ML model on an open-source ECG dataset to predict blood pressure

Field Robotics & Research Experience

Graduate Research Assistant, Marine Robotics Lab — University of British Columbia | Sept 2025 – Present

- Built and field-deployed an underwater ROV for sea urchin population monitoring, working in partnership with the Sc'ianew First Nation
- Designed, developed and tested a surface vehicle driven by a bio-inspired propulsion mechanism
- Developing a simulator to optimize the topology of a bio-inspired AUV and train its deep reinforcement learning controller
- Developing specialized acoustic models for edge-deployed vessel classification and detection

Research Assistant, Monocopter Avionics — Royal Military College of Canada | May 2020 – Sept 2020

- Designed, programmed and manufactured flight avionics for a 375 g single-bladed helicopter (monocopter), culminating in a successful test flight
- Developed a Bluetooth data-acquisition link for real-time monitoring of the vehicle
- Automated wind-tunnel data acquisition with a custom script, saving 2 hours of labour per run and eliminating operator error

Machine Learning Research Assistant — University of British Columbia | Nov 2024 – May 2025

- Built explainable ML models with UBC's urology lab to predict negative outcomes after kidney transplants

Graduate Academic Assistant – Software Engineer — ethicsAI, UBC | Aug 2026 – Present

- Developing an LLM-powered RAG application for teaching ethics to engineering students

Industry Experience

Senior Software Engineer — Key Living Inc. | Dec 2023 – Aug 2025

- Sole software developer, responsible for maintaining and growing the company's backend
- Integrated a publish/subscribe messaging service on Google Cloud, cutting endpoint request times by up to 80%
- Led a testing overhaul that raised coverage by 10%, cut total test run time by 60% and reduced heap usage by 75%

Freelance Software Developer — Self-employed | Jan 2023 – Dec 2023

- Primary developer for a B2C SaaS product: halved its lines of code while improving reliability, and cut image-processing read–write operations by 20%
- Founded and built Comment Explorer, an LLM-powered comment-section analysis app, growing it from 0 to 85 users

Intermediate Software Developer — Properly Inc. | Sept 2021 – Nov 2022

- Led the integration of TypeScript into the codebase, writing standards and supporting its company-wide adoption
- Organized and co-chaired cross-team meetings that launched multiple refactoring initiatives
- Built a notification management system that eliminated support requests about notification settings

Junior Member of Technical Staff – Software Development — MDA (MacDonald, Dettwiler and Associates) | Jan 2021 – Sept 2021

- Developed high-reliability control software for a centrifuge on the International Space Station
- Developed the GUI used to operate the centrifuge reliably
- Supported design of the ground and flight network architecture for Canadarm3 on the Lunar Gateway

Awards

- **J K Zee Memorial Fellowship**, UBC — $20,000 (2026)
- **UBC Student Fellowship** — $62,000 over two years (2025 – Present)
- **Faculty of Applied Science Graduate Award**, UBC — $1,500 (2026)
- **Research presentation awards (2026):** 2nd, UBC ECE Research Day; 2nd, UBC Three Minute Thesis; 3rd, SNAME Vancouver Student Presentation Night

Leadership

- **VP Academic**, ECE Graduate Student Association, UBC (May 2026 – Present) — lead academic event planning
- **PRISMS Student Liaison**, Pacific Rim Initiative for Sustainable Marine Systems, UBC (June 2026 – Present) — organize events for UBC's marine sciences cluster